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TASK 5: UI DESIGN AND IMPLEMENTATION

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1. INTRODUCTION

A. FOREWORD

In the current digital era, user interface (UI) design is crucial in shaping user experiences and the success of software applications. Whether it's a mobile app, a web platform, or a desktop program, the UI acts as a vital link between users and technology, enabling interaction and improving usability.

This report on UI design and implementation aims to provide a thorough overview of the principles, methodologies, and best practices involved in creating intuitive and engaging user interfaces. It examines the essential elements of UI design, from understanding user needs and behaviors to developing visually appealing and functional interfaces. Moreover, it covers the implementation stage, where design concepts are transformed into interactive and responsive user experiences.

The insights presented in this report are based on both theoretical knowledge and practical experience, offering valuable perspectives for designers, developers, and stakeholders in the UI design process. By addressing the challenges and opportunities in UI design, this report aims to advance user-centered design practices and the development of digital products that are not only visually attractive but also highly functional and accessible.

As you delve into this report, we hope you gain a deeper understanding of the art and science of UI design and find inspiration to apply these principles in your projects. Ultimately, the goal is to create user interfaces that enhance satisfaction, engagement, and productivity for users worldwide.

B. RELATION TO SCOPE

In a fingerprint attendance recording application, the user interface (UI) design and implementation phases are critical to ensuring a seamless, efficient, and user-friendly experience. The success of such an application relies heavily on intuitive design and robust implementation, making these phases crucial to its overall functionality and user adoption. Below are a few steps used in designing it.

2. UI DESIGN PHASE

a) User Research and Requirements Gathering:

- Understand the needs and behaviors of the end-users, including administrators and users.
- Identify key functionalities, such as fingerprint enrollment, time logging, and report generation.

b) Wireframing and Prototyping:

- Create wireframes to outline the basic layout and navigation structure.
- Develop interactive prototypes to visualize the workflow and get early feedback from stakeholders.

c) Visual Design:

- Design a clean and intuitive interface with easy-to-navigate menus and clear instructions for fingerprint scanning.
- Ensure the design is accessible, considering different user demographics and potential accessibility needs.

d) Usability Testing:

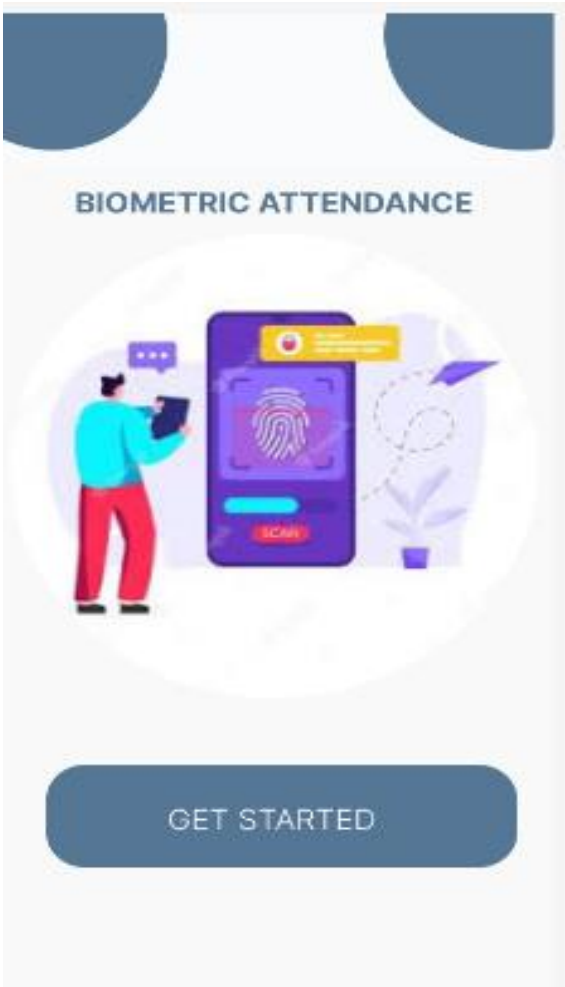
- Conduct usability tests with real users to identify any pain points or areas of confusion.
- Iterate on the design based on feedback to enhance usability and user satisfaction.

2.1. DIFFERENT PAGES

In our project, Figma software was used to design our UI. This entailed designing the various login pages' interfaces for most tasks to be carried out. The order in which the pages will come is as follows:

a) The Welcome/Onboarding Page

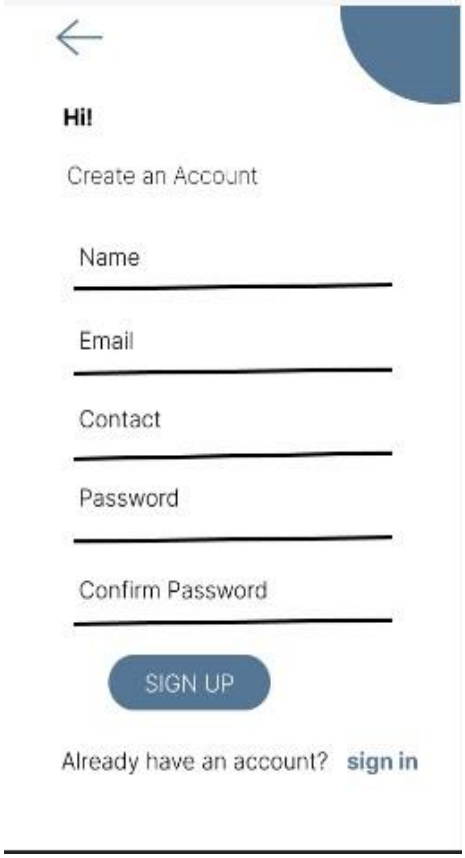
This is the first page that appears when the application is opened. It displays a welcome message and a "Get Started" button. Clicking on the button takes the user to the login page.



b) The Sign-Up Page

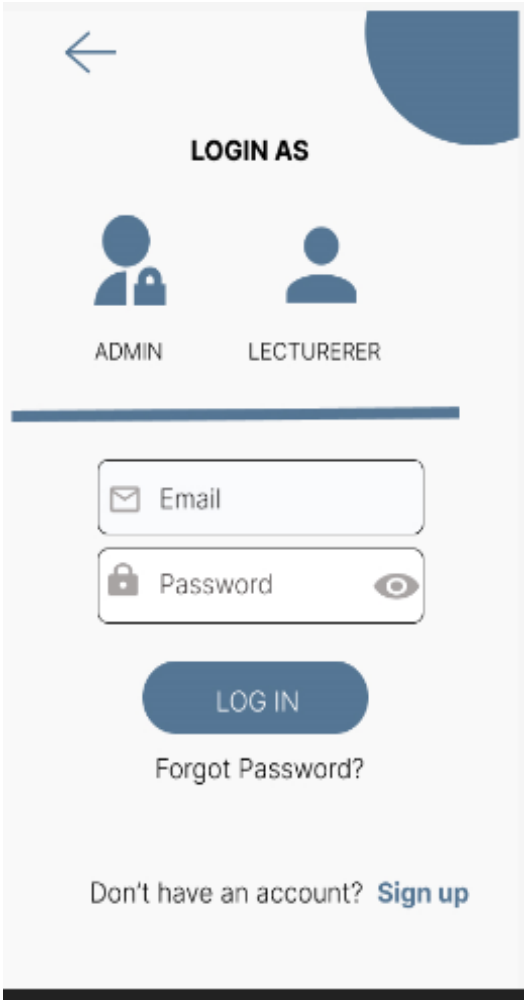
This page is for new users who don't have an account yet. To access this page, users should click on the "Don't have an account yet? Sign up" option on the login page. On the sign-up page:

- Users are required to enter their full names, which must match the records in the school database but not be present in the lecturer's database.
- Users must enter a valid email address, their contact information, create a new password, and confirm the password.
- If the user's name already exists in the application's records or is not found in the school's lecturer database, an error message will be displayed, and the user will have to start the process over.
- If the account is successfully created, the user is redirected to the login page, where they can enter their credentials and wait for the administrator to validate their account.



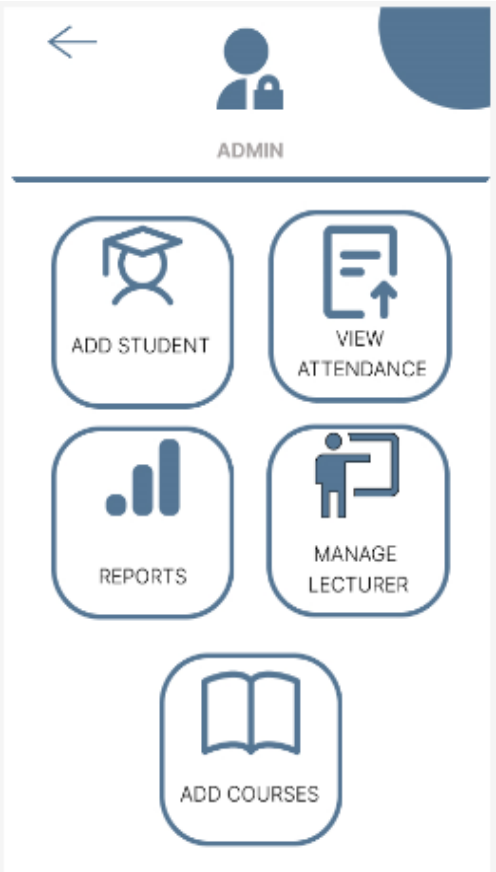
c) The Login Page

This is the first page that appears after the welcome page. On this page, users must enter their email address and password. They are then prompted to choose whether they are logging in as an admin or a lecturer. The following occur based on the user's choice:



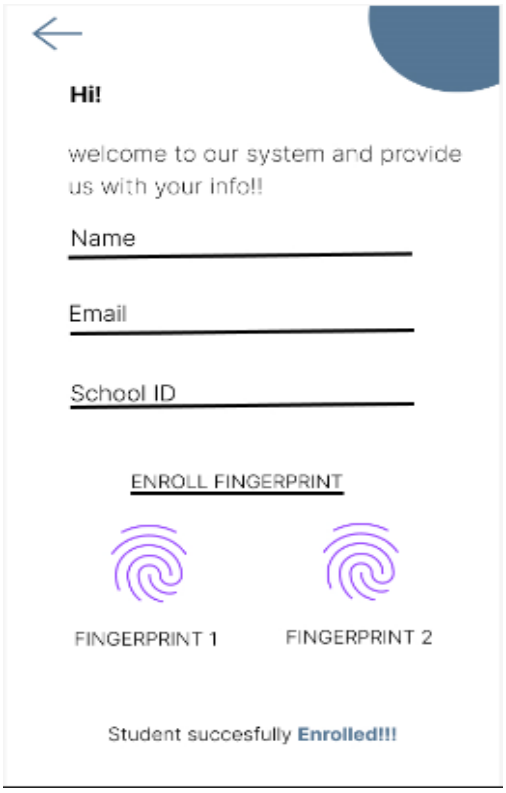
d) Admin Account (Dashboard)

This account opens if an admin account with the entered credentials exists. The admin dashboard provides access to various functionalities such as adding students, viewing attendance, managing lecturers, generating reports, and adding courses. The following pages and actions are available based on the selected option:



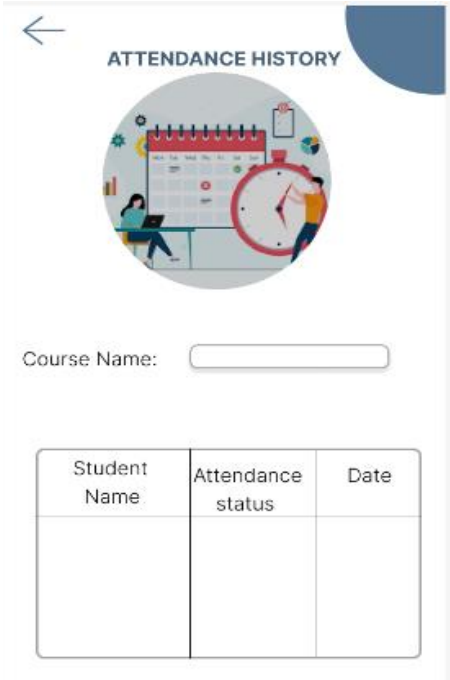
e) Add Student Page

- Admins can add students by entering their names, email addresses, and school IDs (matriculation numbers).
- The system verifies the entered information against the student records database.
- Once verified, students can enter their fingerprints twice for accuracy. The fingerprint data is then hashed and stored in the database, completing the student registration process.



f) View Attendance Page:

- Admins can view attendance history for various courses by entering a valid course name and date.
- An attendance record is generated and displayed in a tabular format with columns for student names and attendance status.

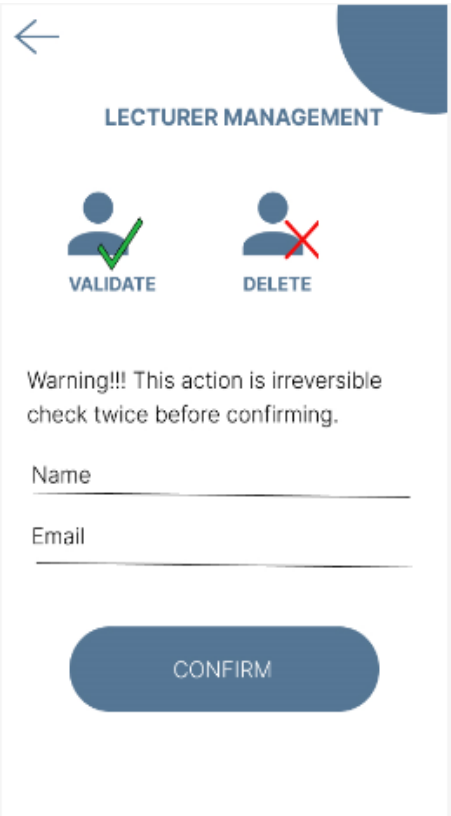


g) Reports Page

This page generates general attendance reports and provides statistical data.

h) Manage Lecturer Page

- Validate Lecturer's Account: Enter the lecturer's name and email address to validate their account.
- Delete Lecturer: Enter the lecturer's name and email address, along with a reason for deletion. The account is suspended while the reason is reviewed by the board of administrators and deleted if validated.

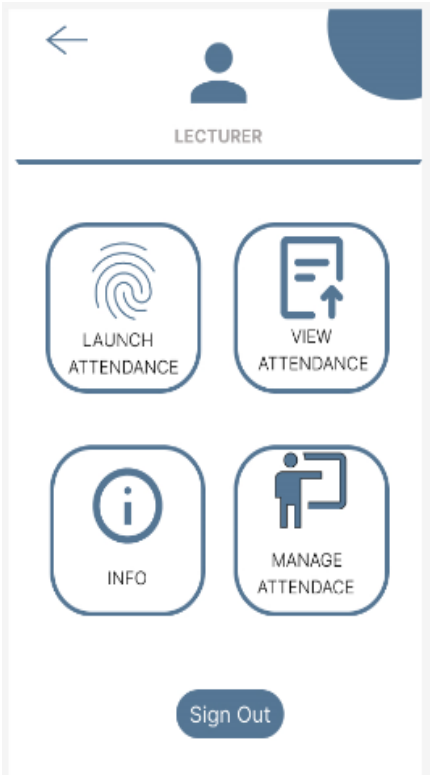


i) Add Courses Page

- Admins can add courses by entering the course name, course code, lecturer's name, and department.
- Upon successful entry, the course is added to the system.

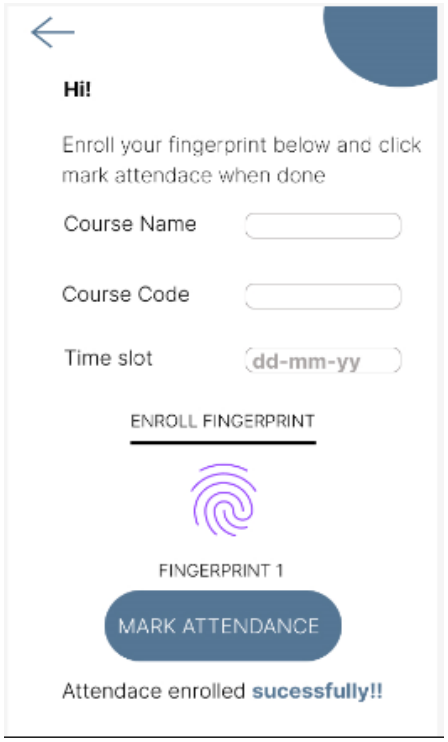
j) Lecturer Account (Dashboard)

Lecturers can access the following pages: launch attendance, view attendance, info, manage attendance, and sign out. Each page is detailed below:



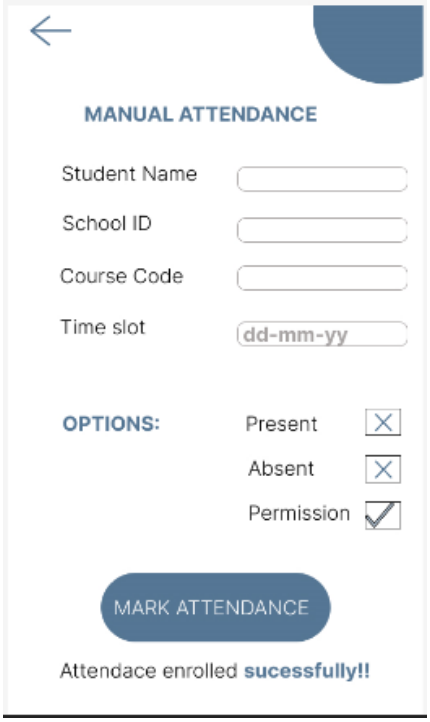
k) Launch Attendance

- Lecturers initiate attendance by entering the course title, course code, and date.
- The system determines the class period automatically. Lecturers then click "Enroll Fingerprint" to blur and close the menu.
- Students mark their attendance by clicking "Mark Attendance" and entering their fingerprints in the biometric sensor.



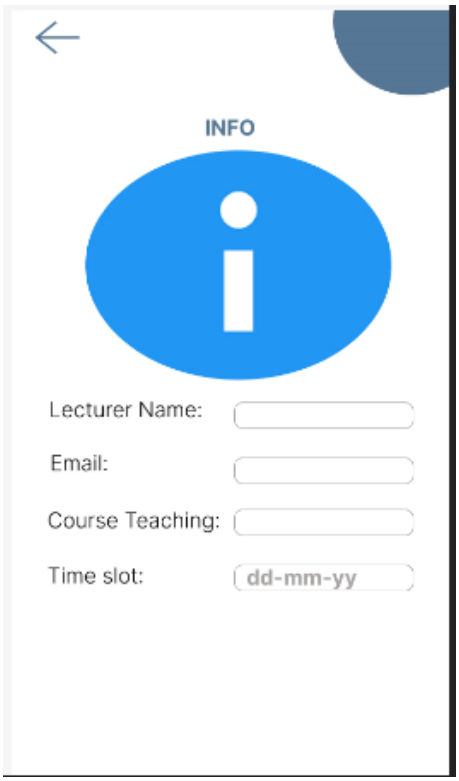
l) Manual Attendance

In cases of incorrect fingerprint entry, justified absence, or when a student is asked to leave, lecturers can manually mark students as present, absent, or granted permission.



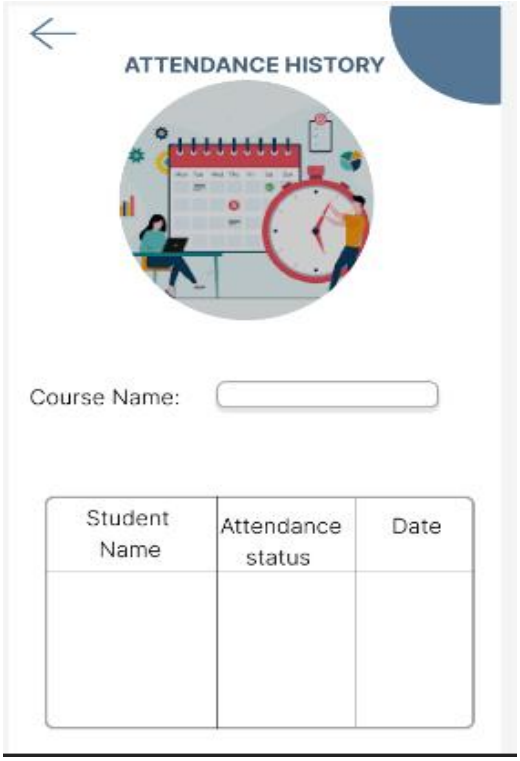
m) Info Page

This page displays the lecturer's information, including name, email address, courses taught, and class schedules.



n) View Attendance

Lecturers can generate attendance records for a specific period. This page mirrors the Attendance History Page in the Admin Account.



3. CONCLUSION

This design leverages modern design principles and best practices to create a cohesive and aesthetically pleasing user experience. It is responsive and adaptable, ensuring consistent performance across various devices, including tablets and smartphones.

By integrating biometric authentication, the app not only improves attendance accuracy but also enhances security, making it a robust solution for educational institutions. The clean and intuitive interface reduces the learning curve for users, promoting widespread adoption and satisfaction.